# 0.3 Report Template – Ship Design Project

Ship and Ocean Systems Design 2026 · Instituto Superior Técnico

| Group | Baseline | Students (name, number) | Specializations | Date |
|---|---|---|---|---|
| [Group n.º] | [A Azores Express / B Azores Feeder] | [names and numbers] | [two specializations] | [date] |

*How to use this template: keep the numbered sections, which follow the ten deliverables D1–D10 of 0.2 Ship Design Project. Replace every text in brackets and delete these italic notes. Hand in the report, the datasheet and your files in one ZIP by e-mail, by 13 December 2026. Declare any use of AI tools in section 0.*

## 0 Summary

*Half a page: the ship, its main particulars, the design objective and how well the design meets it. Declare here the AI tools used and what for.*

## D1 Design Basis

*Must show: owner's requirements, port limits, environmental targets. Use the method of Exercises 1.2 and 1.3 (practised on the reference ship) for your own ship, with every requirement tagged and sourced.*

| Requirement | Value | Unit | Source | Status (Given / Assumed / To be checked) |
|---|---|---|---|---|
| [Capacity] | [ ] | [TEU] | [0.2] | [ ] |
| [Service speed] | [ ] | [kn] | [ ] | [ ] |
| [Port draught limit, each port] | [ ] | [m] | [ ] | [ ] |

# D2 Reference Ship Database

*Must show: recent similar ships, their sources, and the regressions with their bounds.*

| Ship | Year | TEU | DWT [t] | LOA [m] | LPP [m] | B [m] | D [m] | T [m] | V [kn] | P [kW] | Source |
|---|---|---|---|---|---|---|---|---|---|---|---|
| [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] | [ ] |

*For each regression: the equation, $R^2$, the standard error and the range of validity. Plot every regression with your ship on it.*

# D3 Ship Model

*Must show: inputs and outputs, the calculation sequence, and a closed weight balance (Δ = lightship + DWT, within your tolerance).*

# D4 Voyage and Financial Model

*Must show: ship model outputs used as inputs, port restrictions, revenue assumptions. Round voyage or loop time, fuel or energy per voyage, costs, required freight rate and NPV.*

# D5 Validation Note

*Must show: the model run for reference ships, the error per quantity, and how the model was corrected.*

| Quantity | Reference ship | Model | Error [%] | Action |
|---|---|---|---|---|
| [Lightship] | [ ] | [ ] | [ ] | [ ] |

# D6 Optimization Report

*Must show: a justified objective, the design variables and their bounds, the active constraints, and the optimum compared with the requirements.*

# D7 Compliance Sheet

*Must show: attained EEDI against required and CII rating, or not applicable, with the reason; energy per TEU-mile (kWh/TEU·nm) for the battery-electric Feeder; freeboard and intact stability, draught at every port of call.*

| Check | Required | Attained | Pass |
|---|---|---|---|
| [EEDI] | [ ] | [ ] | [yes / no / not applicable, with the reason] |
| [Draught at each port] | [ ] | [ ] | [ ] |

# D8 General Arrangement and Hull Form

*Must show: an A3 sketch with profile and midship section, and a hydrostatics check of the hull form against the ship model.*

# D9 Specialization Documents

*One subsection per specialization, as its self-study deck asks: calculation note, required drawings, acceptance checks.*

## 9.1 [Specialization 1]

## 9.2 [Specialization 2]

# D10 Datasheet and Presentation

*Attach the one-page A4 datasheet (0.4 Datasheet Template). The 15-minute presentation on 18 December follows the same order as this report.*

# References

*Author (year), title, publisher or journal. Cite Papanikolaou (2026) by section.*

# Appendices

*Code, spreadsheets, and the full reference ship data, as files in the ZIP.*